Balance Table

Covariate balance and wave composition (companion to Causal Forest.Rmd)

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Balance checks for the causal forest sample. They do not depend on any forest, so they live here rather than in `Causal Forest.Rmd`. The sample and cleaning steps below are identical to the ones in that file.

1 Setup and data preparation

```
required_packages <- c("haven", "estimatr", "dplyr", "knitr")
missing_packages  <- setdiff(required_packages, rownames(installed.packages()))
if (length(missing_packages) > 0) {
  install.packages(missing_packages, repos = "https://cloud.r-project.org")
}
library(haven)
library(estimatr)
library(dplyr)
library(knitr)
```

```
data_path <- file.path("data", "oreopoulos-resume-study-replication-data-file.dta")
if (!file.exists(data_path)) {
  stop("Data file not found. Expected it at: ",
       normalizePath(data_path, mustWork = FALSE))
}
df <- read_dta(data_path)
```

```
# Pre-treatment resume characteristics
all_pretreat_cov <- c(
  "language_skills",
  "female",
  "ba_quality",
```

```
  "extracurricular_skills",
  "ma",
  "same_exp",
  "exp_highquality"
)

# Keep Canadian-named (control) and Chinese/Indian/Pakistani-named (treated) resumes.
# Characteristics that are missing in some waves are recoded to 0.
# NB: same_exp is missing for every resume in wave 2, so it is 0 for all of wave 2.
# reference, accreditation and legal are only observed in wave 2, so they are not used.
df_int1 <- df %>%
  filter(
    type %in% c(0, 1),
    name_ethnicity %in% c("Canada", "Chinese", "Indian", "Pakistani")
  ) %>%
  mutate(
    # D: 1 = foreign name (type 1), 0 = Canadian name (type 0)
    D = as.numeric(type == 1),
    across(all_of(all_pretreat_cov), ~ ifelse(is.na(.x), 0, .x))
  )

waves <- sort(unique(df_int1$fall_data))
```

2 Covariate balance table

SMD = (mean in D=1 minus mean in D=0) / sqrt((var in D=1 + var in D=0) / 2). Rule of thumb: |SMD| < 0.1 counts as balanced. The p-value is from a firm-clustered regression of the variable on D. Wave is a three-level code, so it enters as one dummy per wave rather than as a numeric code.

```
smd <- function(x, d) {
  (mean(x[d == 1]) - mean(x[d == 0])) /
    sqrt((var(x[d == 1]) + var(x[d == 0])) / 2)
}

# One dummy per wave
df_bal <- df_int1
for (k in waves) df_bal[[paste0("wave_", k)]] <- as.numeric(df_bal$fall_data == k)

bal_vars <- c(all_pretreat_cov, paste0("wave_", waves))

balance_row <- function(v, dat) {
  x <- as.numeric(dat[[v]])
  p <- tryCatch({
    fit <- lm_robust(as.formula(paste(v, "~ D")), data = dat,
                     clusters = firmid, se_type = "CR2")
    unname(fit$p.value["D"])
  }, error = function(e) NA_real_)
  # Percentages only make sense for 0/1 variables
  is_bin <- all(x %in% c(0, 1))
  data.frame(variable = v,
             mean_D0  = mean(x[dat$D == 0]),
             mean_D1  = mean(x[dat$D == 1]),
```

```
            pct_D0   = if (is_bin) 100 * mean(x[dat$D == 0]) else NA_real_,
            pct_D1   = if (is_bin) 100 * mean(x[dat$D == 1]) else NA_real_,
            SMD      = smd(x, dat$D),
            p_clustered = p)
}

bal_tab <- bind_rows(lapply(bal_vars, balance_row, dat = df_bal)) %>%
  mutate(flag = ifelse(abs(SMD) >= 0.1, "check", ""))

kable(bal_tab, digits = 3,
      col.names = c("variable", "mean D=0", "mean D=1", "% D=0", "% D=1",
                    "SMD", "p (clustered)", "flag"),
      caption = paste("Balance of resume characteristics by name type",
                      "(D = 1 foreign name); % columns shown for 0/1 variables"))
```

Table 1: Balance of resume characteristics by name type (D = 1 foreign name); % columns shown for 0/1 variables

variable	mean D=0	mean D=1	% D=0	% D=1	SMD	p (clustered)	flag
language_skills	0.247	0.257	24.719	25.694	0.022	0.384	
female	0.510	0.506	51.024	50.627	-0.008	0.767	
ba_quality	0.590	0.636	58.956	63.588	0.095	0.000	
extracurricular_skills	0.600	0.608	60.046	60.813	0.016	0.593	
ma	0.205	0.198	20.489	19.840	-0.016	0.547	
same_exp	0.284	0.243	28.387	24.287	-0.093	0.000	
exp_highquality	0.331	0.318	33.080	31.775	-0.028	0.328	
wave_0	0.315	0.361	31.494	36.070	0.097	0.000	
wave_1	0.340	0.177	34.005	17.712	-0.379	0.000	check
wave_2	0.345	0.462	34.501	46.218	0.241	0.000	check

3 Wave checks

The balance table shows treated share differs a lot by wave. These two checks ask whether that is a composition issue (which name types were sampled in which wave) or a randomisation problem within wave. (Whether the name penalty itself differs by wave is checked in `Causal Forest.Rmd`.)

3.1 Wave composition

```
# Which name types appear in which wave?
print(table(wave = df_int1$fall_data,
            ethnicity = as.character(df_int1$name_ethnicity)))
```

```
##     ethnicity
## wave Canada Chinese Indian Pakistani
##    0    953     481    468         0
##    1   1029       0      0       466
##    2   1044     517    699         0
```

```r
# Treatment share and callback rate by wave
wave_tab <- df_int1 %>%
  group_by(wave = fall_data) %>%
  summarise(n = n(),
            treated_share    = mean(D),
            callback_rate_D0 = mean(as.numeric(callback)[D == 0]),
            callback_rate_D1 = mean(as.numeric(callback)[D == 1]),
            .groups = "drop")

kable(wave_tab, digits = 3, caption = "Treatment share and callback rate by wave")
```

Table 2: Treatment share and callback rate by wave

wave	n	treated_share	callback_rate_D0	callback_rate_D1
0	1902	0.499	0.158	0.115
1	1495	0.312	0.127	0.079
2	2260	0.538	0.141	0.098

3.2 Balance within wave

```r
# Difference between name types after absorbing wave fixed effects
within_row <- function(v) {
  dat   <- df_int1
  dat$x <- as.numeric(dat[[v]])
  fit   <- lm_robust(x ~ D + factor(fall_data), data = dat,
                     clusters = firmid, se_type = "CR2")
  data.frame(variable         = v,
             diff_D1_minus_D0 = unname(coef(fit)["D"]),
             se               = unname(fit$std.error["D"]),
             p_clustered      = unname(fit$p.value["D"]))
}

within_tab <- bind_rows(lapply(all_pretreat_cov, within_row))

kable(within_tab, digits = 3,
      caption = paste("Balance within wave (name-type difference,",
                      "wave fixed effects, firm-clustered)"))
```

Table 3: Balance within wave (name-type difference, wave fixed effects, firm-clustered)

variable	diff_D1_minus_D0	se	p_clustered
language_skills	-0.003	0.011	0.811
female	-0.001	0.014	0.963
ba_quality	0.015	0.012	0.227
extracurricular_skills	0.007	0.015	0.609
ma	-0.008	0.011	0.468
same_exp	0.000	0.009	0.977
exp_highquality	0.010	0.014	0.473